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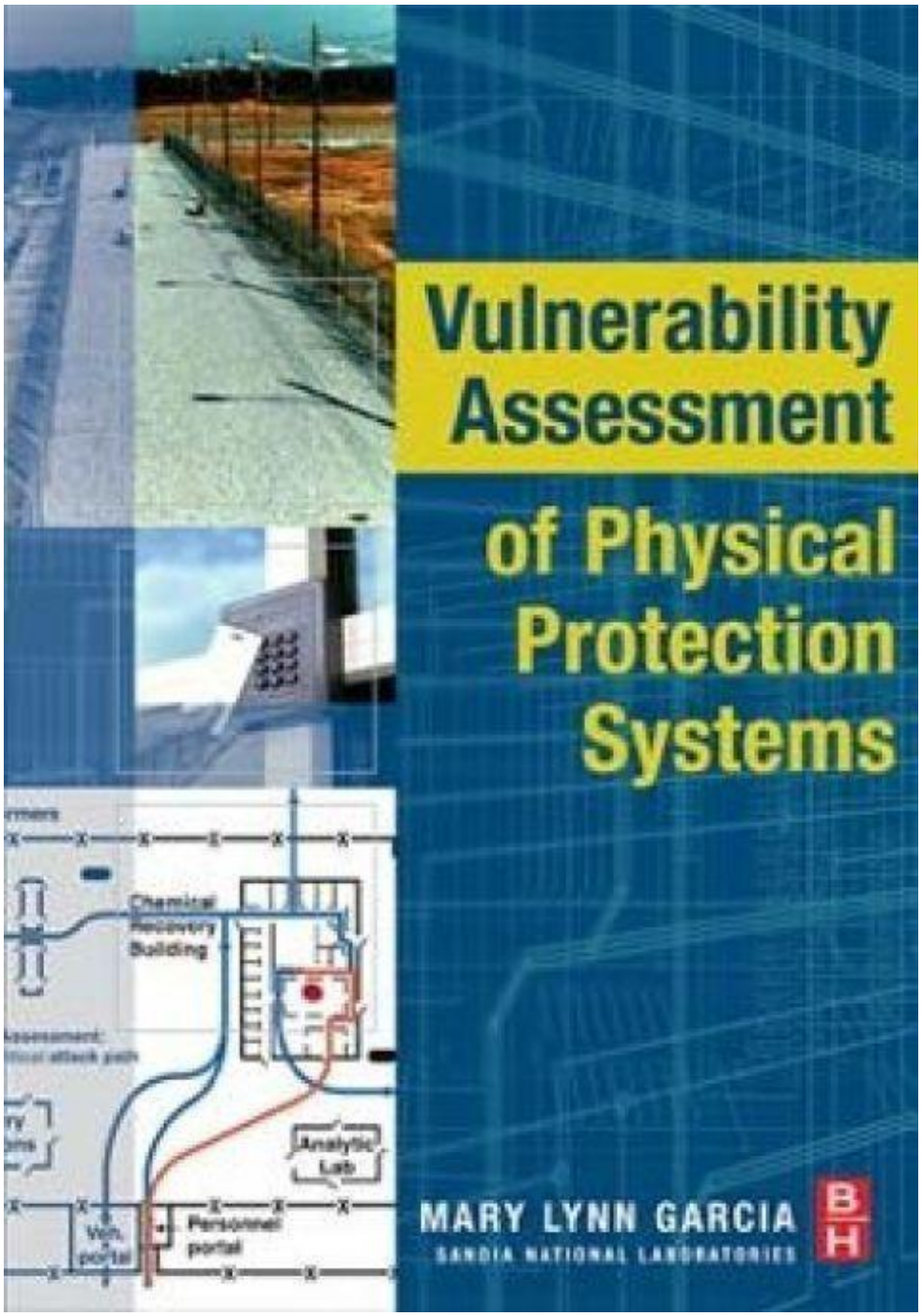

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Vulnerability Assessment of Physical Protection Systems

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For all those responsible for protecting our nation's assets

The search for static security, in the law and elsewhere, is misguided. The fact is, security can only be achieved through constant change, adapting old ideas that have outlived their usefulness to current facts.

William O. Douglas, U.S. Supreme Court justice

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Preface

This book is an extension of the *Design and Evaluation of Physical Protection Systems* text released in April 2001. That book introduced the overall process and principles of physical protection systems; this one describes the application of that process and principles to vulnerability assessment. As in the previous text, the underlying premise of this book is that overall system effectiveness is the objective of a physical security system, not merely the presence of nonintegrated equipment, personnel, and procedures. There is a need for such a text, despite the presence of many existing practices and publications that address vulnerability assessment. Many of these practices and publications rely on compliance-based approaches to security, even for critical assets. This is further constrained by the need to justify the business case for security to senior managers. The fundamental objective of a security system is to protect assets. It is agreed that investments in security should be justified, but protection system effectiveness must also be demonstrated for this investment. Therefore, the simple presence of people, procedures, or equipment does not necessarily provide protection. There is no reason why enterprises cannot have effective security that also complies with regulations or policies.

Because a vulnerability assessment is meant to identify security system weaknesses that can be exploited by malevolent human threats, the difference in performing the evaluation using a checklist that verifies presence of security components (compliance-based) versus an evaluation that shows system effectiveness (performance-based) is significant. The decision concerning which approach is used should be driven by the consequence of loss of the asset. Every enterprise, no matter the size, has critical assets. Critical assets must be protected from adversaries and require performance-based evaluation approaches to ensure effective protection. Low-value assets require less protection and compliance-based approaches are more appropriate in these cases. Since the attacks on 9/11, some progress has been made in improving physical security for critical or other important assets, but much remains to be done.

Some readers will find parts of this text, particularly analysis, to be overly simple in the scenarios and assumptions that are used in the examples. This is a valid point, but because vulnerability assessment is a sensitive process, caution prevails. The goal was to write a thorough but not overly detailed description of common vulnerabilities of physical protection systems. It is regrettable if this disappoints any readers; however, there are reasonable limits to the type of information that should be publicly available. (Managers who bear the responsibility and accountability for security systems may appreciate this thought more than other readers!)

Readers who are unfamiliar with the first book may find it useful to review the process before reading this text. Chapters 1, 14, and 15 should be sufficient to fill in most of the background, although some terms may still be unfamiliar. Chapter 1 in